

Disproof of conjecture 00000000751:
over $\mathbb{Q}_p$, $z \mapsto z/p$ (degree 1) has a nonempty Julia set and $z \mapsto z^2$
(degree 2) an empty one

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1 The conjecture and the reading

Statement (English, verbatim). Definition: The p-adic Fatou decomposition. Conjecture: For iterations of rational functions over $\mathbb{Q}_p$, the Fatou set within the field of uniformly rational functions is completely invariant and open, and the Julia set is nonempty if and only if the degree is ≥ 2 (a complex-p parallel structure). (The Chinese text says the same: for iteration of rational functions over $\mathbb{Q}_p$, the Fatou set “of the field of uniformly rational functions” is a completely invariant open set, and the Julia set is nonempty if and only if the degree is ≥ 2 .)

Reading. The phrase “within the field of uniformly rational functions” has no standard meaning; we read it as referring to the Fatou set of the iterates of a rational function $\varphi \in \mathbb{Q}_p(z)$, defined by a uniform (equicontinuity) condition. The conjecture then says: for every prime p and every nonconstant $\varphi \in \mathbb{Q}_p(z)$,

$$(A) \ F(\varphi) \text{ is open and } \varphi^{-1}(F) = F = \varphi(F), \quad \text{and} \quad (B) \ J(\varphi) \neq \emptyset \iff \deg \varphi \geq 2.$$

We refute the universal statement by refuting (B), in both directions, for every prime p . The dynamics takes place on the classical projective line $\mathbb{P}^1(\mathbb{Q}_p)$ (the text says “over $\mathbb{Q}_p$ ”); we also prove the same results on $\mathbb{P}^1(\mathbb{C}_p)$. We formalize three readings of the Julia set: (J1) the complement of the Fatou set, where a point is Fatou if it has a neighbourhood on which the iterates $\{\varphi^n\}_{n \geq 0}$ are equicontinuous for the chordal metric; (J2) the set of points at which the iterates are not equicontinuous; (J3) the closure of the set of repelling periodic points. In each case the Fatou set in (A) is the complement of the Julia set.

Not covered. We do not treat the Berkovich projective line. The degree-2 example below is a statement about the classical line only. The degree-1 example also concerns the classical line, but it uses no property of maps of degree ≥ 2 .

2 Definitions and sources

Projective line and chordal metric. $\mathbb{P}^1(K) = K \cup \{\infty\}$ (Lean: an inductive type with constructors **aff** $z = [z : 1]$ and **infy** $= [1 : 0]$). Benedetto and Lee, *J-Stability in non-archimedean dynamics*, arXiv:2102.05841, Section 2.1: “Equivalently, in affine coordinates we have” $\rho(z, w) = \frac{|z-w|}{\max\{1, |z|\} \max\{1, |w|\}}$, which equals $|z-w|$ if $z, w \in \mathcal{O}$, $|1/z - 1/w|$ if $z, w \notin \mathcal{O}$, and 1 otherwise; they add “See [Sil07, Section 2.1] or [B19, Section 5.1] for more on the chordal metric.” We use $\rho(z, w) = |z-w|/(M(z)M(w))$ with $M(z) = \max\{1, |z|\}$, $\rho(z, \infty) = 1/M(z)$ and $\rho(\infty, \infty) = 0$.

The value at ∞ agrees with their case formula with $1/\infty = 0$. We prove in Lean that ρ is a metric (triangle inequality included); the topology and the uniform structure of $\mathbb{P}^1(K)$ are the ones it induces.

Fatou and Julia sets. Wikipedia, *Arithmetic dynamics* (https://en.wikipedia.org/wiki/Arithmetic_dynamics): “The metric on K and the standard definition of equicontinuity leads to the usual definition of the Fatou and Julia sets of a rational map”, and “A striking difference is that in the nonarchimedean setting, the Fatou set is always nonempty, but the Julia set may be empty.” Benedetto–Lee, Section 2: the type I Fatou set consists of points “having a neighborhood on which the sequence of iterates” “is equicontinuous with respect to the chordal metric”. A periodic point with multiplier $|\lambda| > 1$ “is said to be repelling”. These definitions and quotes are reused from our package for conjecture 00000000745 and were retrieved again for this package.

Rational functions and degree. $\varphi \in K(z)$ is written P/Q in lowest terms with Q monic (Mathlib `RatFunc.num`, `RatFunc.denom`). Wikipedia, *Rational function* (https://en.wikipedia.org/wiki/Rational_function): “Most commonly, the degree of a rational function is the maximum of the degrees of its constituent polynomials P and Q , when the fraction is reduced to lowest terms.” Action on $\mathbb{P}^1(K)$ (Lean `act`): $\varphi(a) = \infty$ if $Q(a) = 0$, otherwise $P(a)/Q(a)$; $\varphi(\infty)$ is ∞ , 0 , or the ratio of leading coefficients, according as $\deg P > \deg Q$, $\deg P < \deg Q$, or $\deg P = \deg Q$. For a polynomial P of degree ≥ 1 this is $z \mapsto P(z)$, $\infty \mapsto \infty$ (Lean `act_poly`).

Conventions. (i) Equicontinuity is Mathlib’s `EquicontinuousOn` / `EquicontinuousAt` for the family $(\varphi^n)_{n \in \mathbb{N}}$, where φ^0 is the identity. (ii) In (J3), x is periodic if $\varphi^n(x) = x$ for some $n \geq 1$, not necessarily minimal. This only enlarges the set, and for z/p we use $n = 1$. (iii) The multiplier of φ^n at an affine point z is the derivative at z of $w \mapsto \varphi^n(w)$, where the affine coordinate of ∞ is set to 0 ; the derivative only depends on a neighbourhood. At ∞ it is the derivative at 0 of the conjugate $w \mapsto 1/\varphi^n(1/w)$. For $\varphi(z) = z^2$ the n -th iterate is $w \mapsto w^{2^n}$ in both charts. For $\psi(z) = z/p$ the n -th iterate is $w \mapsto w/p^n$ in the affine chart and $w \mapsto p^n w$ in the chart at ∞ ; the repelling fixed point used below is the affine point 0 with $n = 1$, where the relevant map is $w \mapsto w/p$ (proved in Lean: `lin_iterate`, `zero_mem_repelling_lin`). (iv) The field K is any nontrivially normed field with an ultrametric norm; we apply the results to $K = \mathbb{Q}_p$ and $K = \mathbb{C}_p$ (Mathlib `Padic`, `PadicComplex`). The degree-1 part does not use the ultrametric inequality.

3 Proof

Lemma 1 (`tri_aff`, `tri_inf_right`, `tri_inf_left`, `tri_inf_mid`; `MetricSpace` instance). ρ is a metric on $\mathbb{P}^1(K)$.

Proof. For affine x, y, z , $(x - z)y = (x - y)z + (y - z)x$ gives $|x - z||y| \leq |x - y|M(z) + |y - z|M(x)$, and $|x - z| \leq |x - y| + |y - z|$ gives the same bound for $|x - z| \cdot 1$. Hence $|x - z|M(y) \leq |x - y|M(z) + |y - z|M(x)$, which is the triangle inequality after dividing by $M(x)M(y)M(z)$. The cases involving ∞ reduce to $M(y) \leq |x - y| + M(x)$ and $|x - z| \leq M(x) + M(z)$. $\square$

Lemma 2 (`lipschitz_sq`). For $\varphi(z) = z^2$, $\rho(\varphi(a), \varphi(b)) \leq \rho(a, b)$ for all $a, b \in \mathbb{P}^1(K)$.

Proof. $M(x^2) = M(x)^2$. For affine x, y : $|x^2 - y^2| = |x - y||x + y| \leq |x - y|\max\{|x|, |y|\} \leq |x - y|M(x)M(y)$, by the ultrametric inequality, so $\rho(x^2, y^2) \leq |x - y|M(x)M(y)/(M(x)^2M(y)^2) = \rho(x, y)$. Also $\rho(x^2, \infty) = 1/M(x)^2 \leq 1/M(x)$. $\square$

Theorem 3 (`fatouSet_sq`, `juliaSet_sq`, `juliaSetPt_sq`, `repelling_sq`, `juliaRep_sq`). For $\varphi(z) = z^2$ (degree 2, `deg_sqFun`), $F = \mathbb{P}^1(K)$, and the sets (J1), (J2), (J3) are empty.

Proof. All iterates are 1-Lipschitz, so the family is uniformly equicontinuous, which gives (J1) and (J2). For (J3): $\varphi^n(z) = z^{2^n}$. If z is affine and $z^{2^n} = z$ with $n \geq 1$, then $|z| \leq 1$ (otherwise $|z|^{2^n} > |z|$), so the multiplier $2^n z^{2^n-1}$ has $|2^n z^{2^n-1}| \leq 1$, since integers have norm ≤ 1 in an ultrametric field. At ∞ the conjugate is $w \mapsto w^{2^n}$, whose derivative at 0 is 0. So no periodic point is repelling, and the closure of the empty set is empty. $\square$

Theorem 4 (`not_equicontinuousAt_lin`, `zero_mem_juliaSet_lin`, `zero_mem_juliaSetPt_lin`, `zero_mem_repelling_lin`, `zero_mem_juliaRep_lin`). *Let $c \in K$ with $0 < |c| < 1$ and $\psi(z) = c^{-1}z$ (degree 1, `deg_linFun`). Then the point 0 lies in each of (J1), (J2), (J3) for ψ .*

Proof. $\psi^n(z) = c^{-n}z$. Put $x_n = c^n$. Then $\rho(x_n, 0) = |c|^n \rightarrow 0$, but $\rho(\psi^n(0), \psi^n(x_n)) = \rho(0, 1) = 1$. So the iterates are not equicontinuous at 0 (J2). A neighbourhood U of 0 on which they were equicontinuous would give equicontinuity at 0, since $\mathcal{N}_U(0) = \mathcal{N}(0)$ (J1). Finally, $\psi(0) = 0$ and the multiplier is c^{-1} , with $|c^{-1}| > 1$. So 0 is a repelling fixed point and lies in the closure of the repelling periodic points (J3). $\square$

Theorem 5 (`both_directions_fail`, `not_conjecture`, `conjecture751_false`, `conjecture751_false_Cp`). *For every prime p and each reading $J \in \{J1, J2, J3\}$, the conjecture “for all $\varphi \in \mathbb{Q}_p(z)$ with $\deg \varphi \geq 1$: (A) and (B)” is false. Moreover $z \mapsto z/p$ has degree 1 and $J \neq \emptyset$, while $z \mapsto z^2$ has degree 2 and $J = \emptyset$. The same holds with $\mathbb{C}_p$ in place of $\mathbb{Q}_p$.*

Proof. Take $c = p$: $|p|_p = p^{-1} \in (0, 1)$, and $|p|$ is the same in $\mathbb{C}_p$ (`PadicComplex.norm_extends'`). If the conjecture held, applying it to ψ (degree 1, $0 \in J(\psi)$) would give $\deg \psi \geq 2$, a contradiction. $\square$

Clause (A) is not examined: the conjunction already fails at clause (B), and it fails in both directions.

4 Lean formalization

File `lean/Conjecture751/Basic.lean` (Lean 4, Mathlib v4.33.1; its only import is `Mathlib`; 440 lines). Objects: `C751.P1`, `P1.chordal` with its `MetricSpace` instance, `act`, `deg`, `fatouSet`, `juliaSet` (J1), `juliaSetPt` (J2), `inv`, `mult`, `repelling`, `juliaRep` (J3), `Conjecture J`, `sqFun = X^2`, `linFun c = C(c^{-1}) \cdot X`. In ASCII ($\sim$ is $\neg$, $\wedge$ is $\wedge$, $\mathbb{Q}_p$ is $\mathbb{Q}_p$):

```
def Conjecture (J : RatFunc K -> Set (P1 K)) : Prop :=
  forall phi : RatFunc K, 1 <= deg phi ->
    (IsOpen (J phi)^c /\ act phi ^-1' (J phi)^c = (J phi)^c /\
      act phi '' (J phi)^c = (J phi)^c) /\
    ((J phi).Nonempty <-> 2 <= deg phi)

theorem conjecture751_false (p : Nat) [Fact p.Prime] :
  (~ Conjecture (juliaSet (K := Q_p))) /\
  ~ Conjecture (juliaSetPt (K := Q_p)) /\
  ~ Conjecture (juliaRep (K := Q_p))) /\
  (exists phi : RatFunc Q_p, deg phi = 1 /\ (juliaSet phi).Nonempty /\
    (juliaSetPt phi).Nonempty /\ (juliaRep phi).Nonempty) /\
  (exists phi : RatFunc Q_p, deg phi = 2 /\ juliaSet phi = {} /\
    juliaSetPt phi = {} /\ juliaRep phi = {})
```

`conjecture751_false_Cp` states the three negations and the degree-2 example for `RatFunc` over $\mathbb{C}_p$. The metric part (the triangle inequality and the chordal formulas) is our own work. The Fatou/Julia definitions follow the 00000000745 package (GPL-3.0), adapted from $\mathbb{Z}_2$ to $\mathbb{P}^1(K)$.

Axiom audit. `#print axioms` for `C751.conjecture751_false`, `C751.conjecture751_false_Cp` and `C751.both_directions_fail` reports only `[propext, Classical.choice, Quot.sound]`. There is no `sorry`, no `native_decide` and no added axiom. `lake build` succeeds.